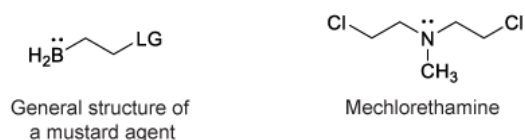


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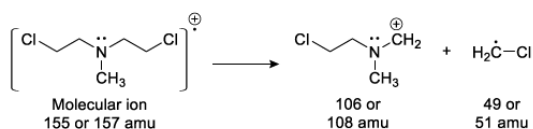
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Reaction 1



In the ^1H NMR spectrum of mechlorethamine, what is the signal area ratio for proton environments found on carbons directly bonded to the nitrogen atom?

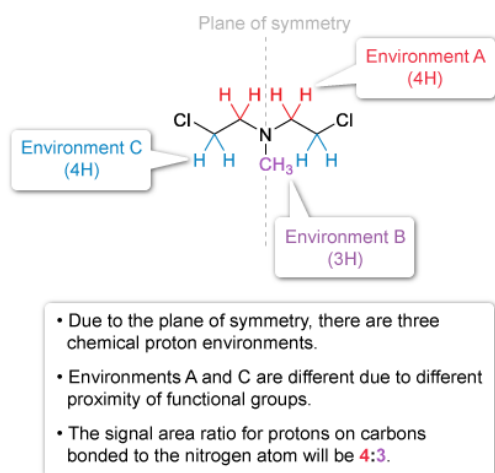
- ✓ ☒ A. 4 to 3 [37%]
- ☐ B. 8 to 3 [8%]
- ☐ C. 2 to 2 to 3 [42%]
- ☐ D. 4 to 4 to 3 [11%]

Correct

37% answered correctly

Explanation:

Proton NMR environments in mechlorethamine



Proton (^1H) nuclear magnetic resonance (NMR) spectroscopy is a laboratory technique used to evaluate molecular structure. Samples are placed in an external magnetic field (B_0) and radio waves induce transitions between low- and high-energy nuclear **spin states**. Relaxation of nuclei to their lower-energy spin state generates signals that indicate:

- The **number of chemical environments**
- The **chemical shift**
- The ratio of signal intensities (**integration**)
- The spin-spin splitting of signals (**multiplicity**)

The chemical environment of a proton is influenced by proximal atoms in the compound's molecular structure. Protons attached to same carbon atom are typically in the same chemical environment. Nearby electronegative atoms or functional groups influence the chemical environment of a proton nucleus. Symmetrical molecules may have fewer unique chemical environments since protons on different carbon atoms may have identical proximity to influencing functional groups.

In a ^1H NMR spectrum, the area of a signal (**integration**) is proportional to the number of protons in the molecule that generated the signal. Therefore, the integration ratio of signals is proportional to the ratio of protons in the respective chemical environments.

Mechlorethamine contains a **tertiary nitrogen**, in which two of the three alkyl groups are identical ($-\text{CH}_2\text{CH}_2\text{Cl}$) due to the plane of symmetry. Consequently, the atoms directly bonded to the nitrogen atom are in one of two chemical environments:

- Environment A: two $-\text{CH}_2-$ groups ($2 \times 2 \text{ H} = 4$)
- Environment B: one $-\text{CH}_3$ group ($1 \times 3 \text{ H} = 3$)

Therefore, the signal area ratio is **4 to 3**.

(Choice B) Although four $\text{-CH}_2\text{-}$ groups are found in the structure of mechloramine, the groups adjacent to the nitrogen (environment B) and adjacent to the chlorine atoms (environment C) are in *different* chemical environments, and therefore appear as two signals.

(Choice C) Due to the plane of symmetry, the two $\text{-CH}_2\text{-}$ groups in environment B have identical chemical environments and generate the *same* ^1H NMR signal, not two signals.

(Choice D) This ratio includes the protons in environment C, but they are *not* attached to a carbon directly bonded to the nitrogen atom, as requested by this question.

Educational objective:

Proton (^1H) NMR spectroscopy is an experimental technique that allows evaluation of molecular structure. Molecules with a plane of symmetry have fewer total chemical environments and signal integration ratios that reflect the symmetry.

Time spent: 0 sec

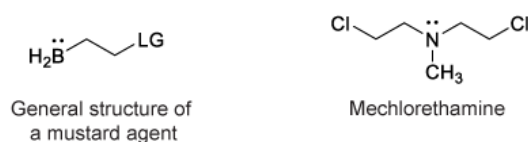
Subject:
Organic Chemistry

QID: 403822

System:
Separation Techniques, Spectroscopy, and Analytical Methods

Mustard agents are a class of highly reactive compounds that cause cytotoxicity and skin damage on contact. After development in Germany during World War I for the purpose of chemical warfare, some nitrogen-containing mustard agents were later found to have medical utility in the treatment of cancer.

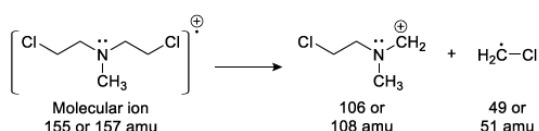
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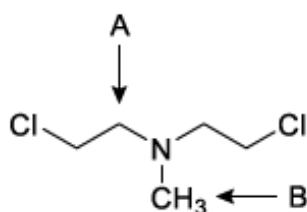
Mechanistically, the nitrogen lone pair undergoes an intramolecular substitution reaction to displace a chloride ion. The resulting cyclic aziridinium intermediate is a potent alkylating agent that reacts with the nucleophilic N7 position of guanine bases in DNA. Since mechlorethamine contains two β -chloro groups, this process can be repeated twice, and the agent is classified as a *bis*-alkylating agent. Mechlorethamine is unstable to basic aqueous conditions, as hydroxide ions irreversibly displace each of its chloride groups and prevent mechlorethamine's ability to be an alkylating agent.

Given the dangers of working with mechlorethamine, it is advantageous to conduct analysis of mechlorethamine samples using spectroscopic techniques that are either nondestructive or require small sample quantities. Techniques such as infrared (IR) spectroscopy, mass spectrometry, and proton (^1H) nuclear magnetic resonance (NMR) spectroscopy have been used to characterize the spectral properties of mechlorethamine. Reaction 1 depicts one possible fragmentation of the mechlorethamine molecular ion during mass spectrometry.

Reaction 1



In the ^1H NMR spectrum of mechlorethamine, which of the following statements most accurately describes the relationship between the chemical shift of proton environment A and proton environment B?



- ✓ ☒ A. Proton environment A has a higher chemical shift than proton environment B. [72%]
☐ B. Proton environment B has a higher chemical shift than proton environment A. [23%]

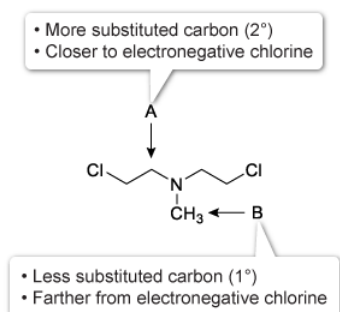
- ☐ C. Proton environments A and B have identical chemical shifts. [3%]
- ☐ D. Nothing can be inferred about the relative chemical shift of proton environments A and B.

Correct

71% answered correctly

Explanation:

Comparison of mechlorethamine chemical shifts



Proton environment A has a higher chemical shift.

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When electrons surrounding a nucleus enter the spectrometer magnetic field (B_0), they generate a magnetic field that opposes B_0 (B_{induced}). The effective magnetic field B_{eff} experienced by a nucleus is impacted by the presence of B_{induced} through a process known as **electron shielding**. **Chemical shifts** (δ) are reported as the ratio of resonance energy frequency (Hz) to the spectrometer frequency (MHz), in units of parts per million (ppm).

Protons in alkyl groups are among the most shielded and normally appear between δ 0.9–1.4, with higher-order (eg, tertiary) alkyl groups having higher chemical shifts than lower-order (eg, primary) alkyl groups. Proton chemical shifts are also influenced by the **inductive effect**, in which nearby electron-rich (ie, electron donating) or electron-deficient (ie, electron withdrawing) groups respectively increase or decrease the effect of shielding.

In this question, the chemical shifts of proton environment A and proton environment B in mechlorethamine are being compared. Since environment B (a primary carbon) is less substituted than environment A (a secondary carbon), the chemical shift of environment A is expected to be higher. In addition, the electronegative chlorine atom is closer to environment A, leading to greater inductive deshielding and a higher chemical shift for environment A. Together, these data point toward the conclusion that **proton environment A has a higher chemical shift than proton environment B**.

(Choice B) Environment B is a less-substituted carbon atom and is farthest from the electronegative chlorine atoms in mechlorethamine. Therefore, environment B has a higher relative electron density and a *lower* chemical shift due to shielding.

(Choice C) Unless two distinct proton environments have identical electronic circumstances, it is exceedingly rare for two environments in the same molecule to have identical chemical shifts. In this question, the two environments have clear differences that lead to a difference in chemical shift.

(Choice D) The structure of mechlorethamine provides sufficient information to predict the relative chemical shift of environments A and B.

Educational objective:

The chemical shift of a ^1H NMR environment is affected by many factors, including the degree of carbon substitution and proximity to electronegative atoms. Using the structure of a molecule, relative chemical shift relationships can be identified.

Time spent: 0 sec

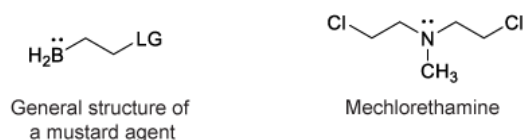
Subject:
Organic Chemistry

QID: 403823

System:
Separation Techniques, Spectroscopy, and Analytical Methods

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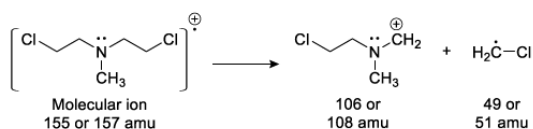
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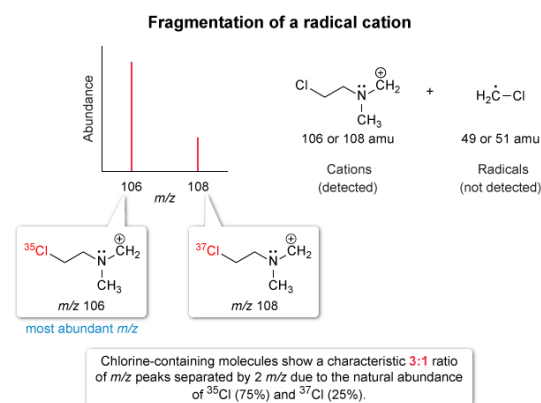
Given the dangers of working with mechlorethamine, it is advantageous to conduct analysis of mechlorethamine samples using spectroscopic techniques that are either nondestructive or require small sample quantities. Techniques such as infrared (IR) spectroscopy, mass spectrometry, and proton (^1H) nuclear magnetic resonance (NMR) spectroscopy have been used to characterize the spectral properties of mechlorethamine. Reaction 1 depicts one possible fragmentation of the mechlorethamine molecular ion during mass spectrometry.

Reaction 1



If Reaction 1 represents the most common process while obtaining a mass spectrum of mechlorethamine, which product mass-to-charge ratio (m/z) peak will have the highest abundance in the compound's mass spectrum? (Note: The isotopic abundances of ^{35}Cl and ^{37}Cl are 75% and 25%, respectively.)

- ☐ A. 49 [16%]
- ☐ B. 51 [11%]
- ☒ C. 106 [56%]
- ☐ D. 108 [15%]

Explanation:

Mass spectrometry provides the ability to measure the mass of molecules and molecular fragments. An **electron-impact mass spectrometer** bombards a sample with electrons to remove an electron from a molecule (**ionization**). This creates the **molecular ion**, a radical cation with a **mass-to-charge ratio (m/z)** equal to the molecular weight of the molecule. The molecular ion can also break into a radical and a cation to produce smaller molecular fragments. Ionized material enters a magnetic field, which curves the path of the particles by their m/z . Only **cations** with m/z values that match the curvature of the instrument tube reach the detector. Finally, a **mass spectrum** (ie, a plot displaying the abundance of detected cations [y-axis] versus m/z [x-axis]) is generated.

Molecules and fragments that contain **isotopes** are detected at different m/z values. The intensity of mass spectrum peaks containing isotopes is determined by the natural isotopic abundance. Although many elements have **isotope effects** that can be observed in a mass spectrum, chlorine and bromine have characteristic isotope signatures.

Reaction 1 shows the fragmentation of the mechlorethamine molecular ion to a cation (m/z 106, 108) and a radical (m/z 49, 51). Each fragment contains one chlorine atom, which has two major isotopes separated by two amu (^{35}Cl and ^{37}Cl). Consequently, each fragment will appear at two m/z values in approximately a 3:1 ratio to reflect the natural abundance of ^{35}Cl to ^{37}Cl .

Since a mass spectrum only detects charged particles (cations), only m/z 106 and m/z 108 will be detected. Of these, the **m/z 106 peak will be the most abundant** given the 75% natural abundance of the smaller ^{35}Cl isotope.

(Choices A and B) A mass spectrometer is only able to detect charged molecules and fragments. The product fragments at 49 and 51 amu are radicals and will *not* be detected.

(Choice D) Although an m/z 108 peak will be detected from this fragmentation, it results from

the less-abundant ^{37}Cl isotope. The abundance of m/z 108 is approximately one third the abundance of m/z 106.

Educational objective:

Mass spectrometry experiments ionize samples and detect the abundance of ions at each mass-to-charge ratio (m/z). Molecules and fragments containing isotopes are detected at

different m/z in ratios that reflect their natural isotopic abundance.

Time spent: 0 sec

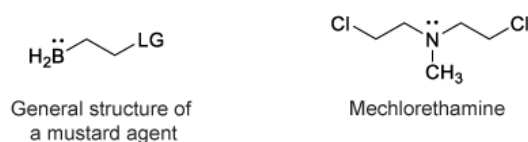
Subject:
Organic Chemistry

QID: 403824

System:
Separation Techniques, Spectroscopy, and Analytical Methods

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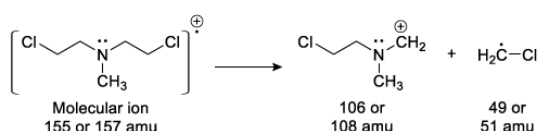
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Reaction 1



Which extraction procedure would be most useful to isolate mechlorethamine from a mixture?

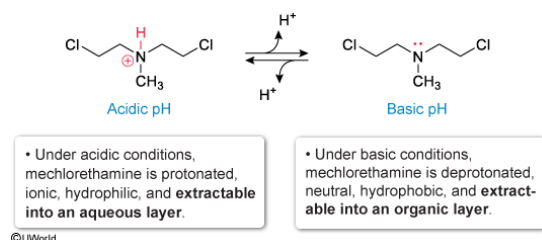
- ☐ A. Extract mechlorethamine into an organic layer from an acidic aqueous layer. [21%]
- ✓ ☒ B. Extract mechlorethamine into an acidic aqueous layer from an organic layer. [42%]
- ☐ C. Extract mechlorethamine into a basic aqueous layer from an organic layer. [20%]
- ☐ D. Extract mechlorethamine into an organic layer from a basic aqueous layer. [15%]

Correct

42% answered correctly

Explanation.

Relationship between acid-base chemistry and extraction



Liquid-liquid extraction uses immiscible solvents (ie, an **organic solvent** and an **aqueous solution**) to separate molecules in a mixture based on **differences in solubility**. The principle "**like dissolves like**" indicates that molecules dissolve in a solvent with a similar polarity. **Hydrophilic molecules** containing polar or ionic bonds have a greater affinity for the polar aqueous layer. Hydrophobic or neutrally charged molecules have fewer polar functional groups, and therefore these molecules have a greater affinity for the nonpolar organic layer. A molecule's charge can be modified to change its affinity for the organic or aqueous layer.

This question asks which extraction procedure would be most useful to isolate mechlorethamine from a mixture. Mechlorethamine contains a basic tertiary amine functional group. Under acidic conditions, the amine lone pair is protonated to generate a hydrophilic ammonium ion. Under basic conditions, the amine exists in its hydrophobic neutral form. By these criteria, mechlorethamine will have the greatest solubility in either:

- An acidic aqueous solution as an ammonium ion (acidic pH)
- An organic solvent as a neutral molecule (basic pH)

Since the passage states that mechlorethamine is unstable under basic conditions, the best choice is to **extract mechlorethamine into an acidic aqueous layer from an organic layer**.

(Choice A) In an acidic aqueous solution, mechlorethamine exists in its ionic ammonium form. As a charged molecule, mechlorethamine has low solubility in the organic layer extract.

(Choice C) To dissolve in an organic layer, mechlorethamine needs to exist in its neutral, hydrophobic form. In this state, mechlorethamine has a greater tendency to remain in the hydrophobic organic layer and is unlikely to be extracted into a basic aqueous layer.

(Choice D) Although neutral, hydrophobic mechlorethamine has greater relative solubility in the organic layer extract, and the passage states that mechlorethamine is unstable under basic aqueous conditions. Therefore, this extraction process is unlikely to result in the extraction of unreacted mechlorethamine.

Educational objective:

Liquid-liquid extraction can be used to separate molecules based on differences in solubility between solutions. Basic functional groups (eg, amines) can be extracted into an acidic aqueous solution through reversible conversion to their ionic, hydrophilic form.

Time spent: 0 sec

Subject:
Organic Chemistry

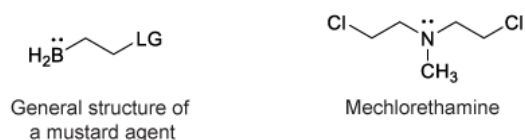
QID: 403825

System:

Separation Techniques, Spectroscopy, and Analytical Methods

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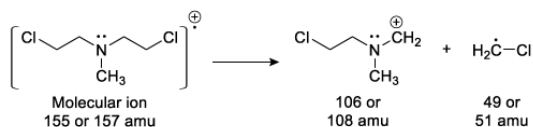
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Reaction 1



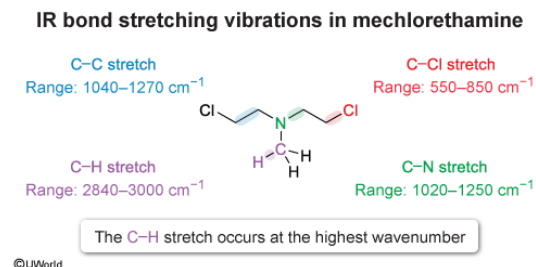
Which of the following stretching vibrations is expected to occur at the highest wavenumber in the infrared spectrum of mechlorethamine?

- ☐ A. The N–C stretch [39%]
- ☐ B. The C–C stretch [5%]
- ☐ C. The C–Cl stretch [30%]
- ✓ ☒ D. The C–H stretch [24%]

Correct

23% answered correctly

Explanation:



Infrared (IR) spectroscopy is an analytical technique in which a sample is irradiated with IR light and a **spectrometer** detects and records the percentage of radiation that passes through the sample over a range of frequencies. Absorption of IR radiation causes **vibrational and rotational motion** in a molecule. Different bonds and functional groups absorb IR radiation at different **intensities** and frequencies, and therefore the signal from each bonding environment appears in a characteristic region of the **IR spectrum**.

The peaks in an IR spectrum are measured in wavenumbers, with units of cm^{-1} . The wavenumber of an IR absorption is proportional to the frequency of the motion and can be approximated by **Hooke's law**. In this model, atoms act as weights on either side of a spring (bond), and heavier atoms lead to slower frequencies of oscillation.

This question asks which of the bond choices will have the highest wavenumber stretching vibration in the IR spectrum of mechlorethamine. As wavenumbers are proportional to frequency, and frequency is inversely proportional to the mass of atoms connected by the bond, the highest wavenumber stretching vibration occurs for the bond with the lightest combined mass of atoms. Of the given choices, C–H is the only bond between a heavy atom and a light hydrogen. Therefore, **the C–H stretching absorption will have the highest wavenumber**.

(Choices A, B, and C) Each of these choices describe a bond between two nonhydrogen atoms. By Hooke's law, the frequency of stretching vibration (and the wavenumber) will be significantly smaller.

Educational objective:

Infrared spectroscopy provides the ability to identify molecular structure through analysis of vibrational and rotational molecular motion. The wavenumber for a stretching absorption can be estimated using Hooke's law.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 403826

System:
Separation Techniques, Spectroscopy, and Analytical Methods